

Traffic Monitoring System

<https://github.com/aiyshwary/Traffic-Monitoring-System>

OVERVIEW

A PHP and SQL-based web application that supports traffic controllers with real-time violator tracking, performance reports, and transport department recommendations for infrastructure improvements.

IMPACT

Developed a multi-role Traffic Monitoring System web application with separate portals for admins, officers, and citizens. Features include real-time violator tracking, fine management, road signalling status, and daily traffic reporting.

KEY DETAILS

1. Admin portal manages user accounts, officer assignments, road data, and signalling configurations.
2. Officer portal logs violators, records fines, and updates real-time road and signal status.
3. Citizen portal supports account registration, violation history lookup, and online fine payment.
4. Automated daily reports summarize traffic violations and flag roads needing infrastructure attention.
5. Built with PHP, MySQL (XAMPP), JavaScript, and CSS with a responsive multi-role front-end.
6. Uses PHP session-based authentication with separate login handlers for citizens and officers, each with unique roles.
7. MySQL database backs five core tables (citizen, officer, fine, road, signalling); citizen signup automatically generates a unique username and Vehicle ID.